

Design of a Modular Digital Data Monitoring Unit with APB4 Bus Integration

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Abstract—Monitoring a sensor often involves software polling and this is not effective in detecting quick spikes. The first phase of the project produced an 8-bit Digital Event Monitoring Unit (DEMU) which has a threshold set by a user, along with a sticky alarm and a fault capture register. However, a unit such as this proves to be quite useless for System-on-Chip because it cannot be configured in any way. The second step of the project takes DEMU and combines it with AMBA APB4 slave interface. As a result, now, the processor can configure the threshold and retrieve the fault that has already been recorded. The information provided in this paper covers the Vivado work on this project, like behavioral simulation, RTL and synthesized schematics, post-implementation utilization, and timing report.

Index Terms—Digital Event Monitoring, Verilog HDL, Sticky Alarm, APB4, SoC Integration, Vivado, Python Automation.

I. INTRODUCTION

IoT sensors and industrial technology sometimes seem to function erratically. However, a processor running software has a chance to miss this voltage anomaly happening for just a few clock cycles. In this case, a hardware FSM watchdog had solved the problem in Phase 1. Unfortunately, the thresholds were to be received from the pins, and a typical SoC just cannot afford to have additional pins for this purpose. In Phase 2, the APB4 wrapper is used to make it possible for a processor to manage the configuration process, find out about faults, and acknowledge alarms without the need for physical pin connections.

II. PROBLEM STATEMENT

A. Phase 1

- 1) When events that are transitory are solely monitored by software, they will be ignored.
- 2) Basic alarms imply nothing about the cause of their triggering; thus, the fault response must be recorded.
- 3) It would not be possible to convert signed comparison to unsigned one during operation.

B. Phase 2

- 1) Hardwired configuration: the pins were used to indicate the threshold values.
- 2) There is no interface of the processor because there is no adopted method for retrieval of fault register data.
- 3) Visualization of design: manually running xilinx to examine the synthesis process takes too long and involves errors.

TABLE I
APB4 MEMORY-MAPPED REGISTER MAP

Offset	Register	R/W	Description
0x00	THRESHOLD	R/W	8-bit threshold value
0x04	MODE_CTRL	R/W	Bit 0 enable, bit 1 data mode
0x08	SW_ACK	W	Write 1 to clear sticky alarm
0x0C	SENSOR_IN	R	Live sensor data
0x10	ALARM_STAT	R	Bit 0 alarm, bits 8:1 fault capture

III. SYSTEM ARCHITECTURE

The design is hierarchical. Phase 1 was a flat core; Phase 2 puts that core one level down inside `apb_demu_wrapper`.

A. Phase 1: DEMU Core

The core (`data_monitor`) has these blocks:

- Sensor and threshold registers that hold the 8-bit values.
- A comparator built from bitwise logic rather than the $<$ and $>$ operators, so the structure is under our control.
- A 3-state Moore FSM: IDLE $\rightarrow$ ALARM $\rightarrow$ COOLDOWN.
- The sticky alarm latch and the fault capture snapshot register.

B. Phase 2: APB4 Wrapper

The wrapper adds four things around the core: the APB handshake logic (`psel`, `penable`, `pwrite`), an address decoder, the register map of Table I, and auto-clear logic that stretches the software acknowledgement into a single-cycle pulse.

IV. SPECIFICATIONS

TABLE II
SYSTEM SPECIFICATIONS (PHASE 1 AND 2)

Parameter	Value
Clock frequency	50 MHz (typical FPGA figure)
Interface	AMBA APB4 (Phase 2), direct pins (Phase 1)
FSM states	IDLE, ALARM, COOLDOWN
Data width	8-bit sensor, 8-bit threshold
Address width	32-bit APB address bus

V. APPLICATIONS AND SDG MAPPING

The same IP drops into very different products: industrial motor protection (SDG 9), automotive battery management (SDG 7), fever detection in medical devices (SDG 3) and flood warning (SDG 11).

VI. IMPLEMENTATION AND RESULTS

A. Phase 1: Core Verification

Eight application testbenches were run against the core. In the waveforms, `ALARM_OUT` stays high and `FAULT_CAPTURE` keeps the trigger value after `SENSOR_DATA` drops back to normal. That is the sticky behavior we wanted.

B. Phase 2: APB4 Behavioral Simulation

A testbench (`tb_apb_wrapper`) uses `apb_write` and `apb_read` tasks to act like a CPU. Fig. 1 shows the run, which ends at 265 ns. Reading it left to right:

- The first write puts `0x50` (80) into address `0x00`, the threshold.
- The second write sets `0x01` at address `0x04`, which turns the monitor on.
- The sensor starts at `0x00`, steps to `0x32` (50) and then jumps to `0x5A` (90). Ninety is above the threshold of 80, so `alarm_output` goes high a little after 140 ns.
- The read at `0x10` comes back with `0x00B5`: alarm bit set, fault value 90 in bits 8:1.
- The last write puts `0x01` into `0x08`, the acknowledge register, and the alarm drops shortly after.

Throughout the run `pready` stays at 1 and `pslverr` stays at 0, so every access completes in the minimum two-phase APB sequence with no error responses.

C. RTL and Synthesized Schematics

Fig. 2 is the elaborated RTL view. The `data_monitor` core (`demu_core_inst`) is the large box in the middle; the `RTL_ROM`, `RTL_MUX`, AND/OR gates, the two subtractors and the asynchronous registers are all visible, and the `prdata` read mux and the address-decoded `sw_ack` mux sit outside it in the wrapper. After synthesis (Fig. 3) the same logic becomes LUT2 to LUT6 cells, four CARRY4 chains for the two difference calculations, FDCE/FDPE flip-flops (the one-hot FSM state uses one FDPE and two FDCE) and an IBUF on every input, with a single BUFG on `pclk`.

D. Utilization and Timing

The post-synthesis utilization is shown in Fig. 6. The whole wrapper takes 52 slice LUTs and 23 slice registers; the core alone is 47 LUTs and 12 registers, so the APB logic adds only about five LUTs and eleven registers on top. Against a device with 14,600 LUTs and 29,200 registers that is under 1% of either. The wrapper uses 88 of 150 bonded IOBs (about 59%), mostly the 32-bit address and write-data buses, and one of 32 BUFGCTRLs.

The timing summary in Fig. 7 reports a worst negative slack of 5.022 ns, a worst hold slack of 0.143 ns and a worst pulse width slack of 4.500 ns. There are no failing endpoints (12 setup/hold endpoints, 24 pulse width endpoints) and Vivado states that all user-specified constraints in `timing.xdc` are met. The hold margin is thin compared with the setup margin, and 0.143 ns is worth keeping an eye on if the design is retimed or placed in a bigger system.

VII. CONCLUSION

Phase 1's independent monitor is now able to communicate using APB4, allowing a processor to determine the threshold, identify what triggered the alarm, and reset it. The simulation results are consistent with the register map, and the implementation is small in size (with 52 LUTs and 23 registers) with low worst-case propagations. Next phases of the project include a moving average filter for sensor output for noisy sensors and switching from APB4 to AXI4-Lite for when bandwidth becomes a problem.

REFERENCES

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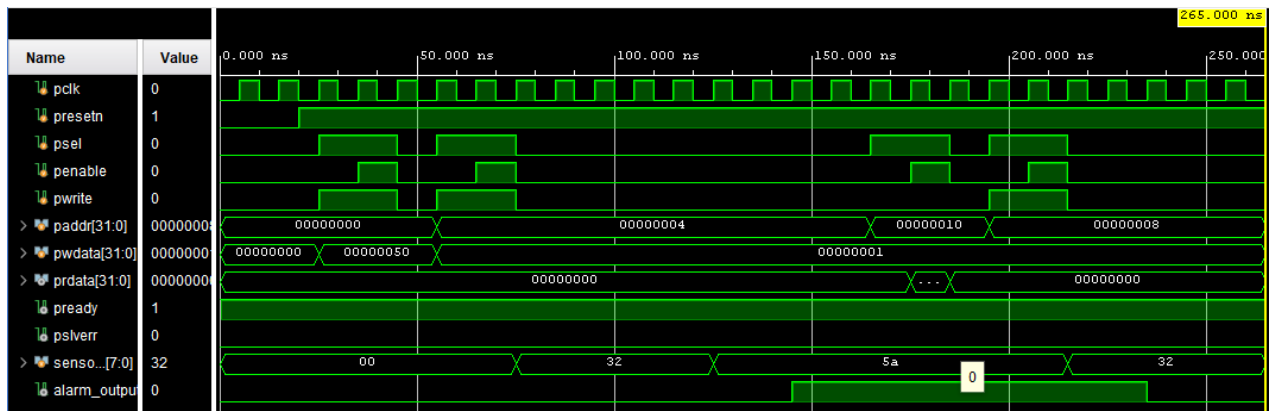


Fig. 1. Behavioral simulation of tb_apb_wrapper in Vivado (0-265 ns).

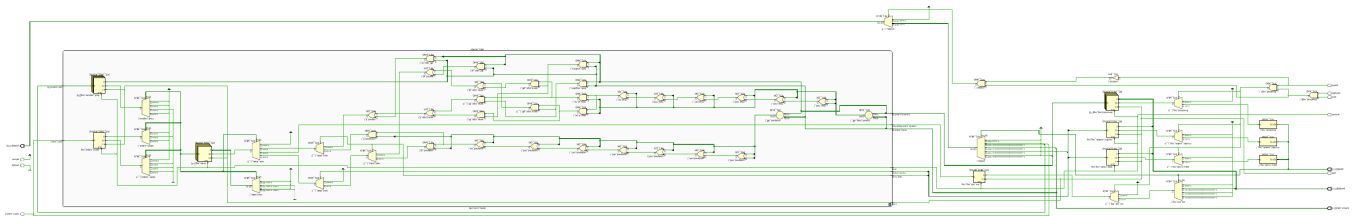


Fig. 2. Elaborated RTL schematic of apb_demu_wrapper, cropped from the Vivado schematic for readability.

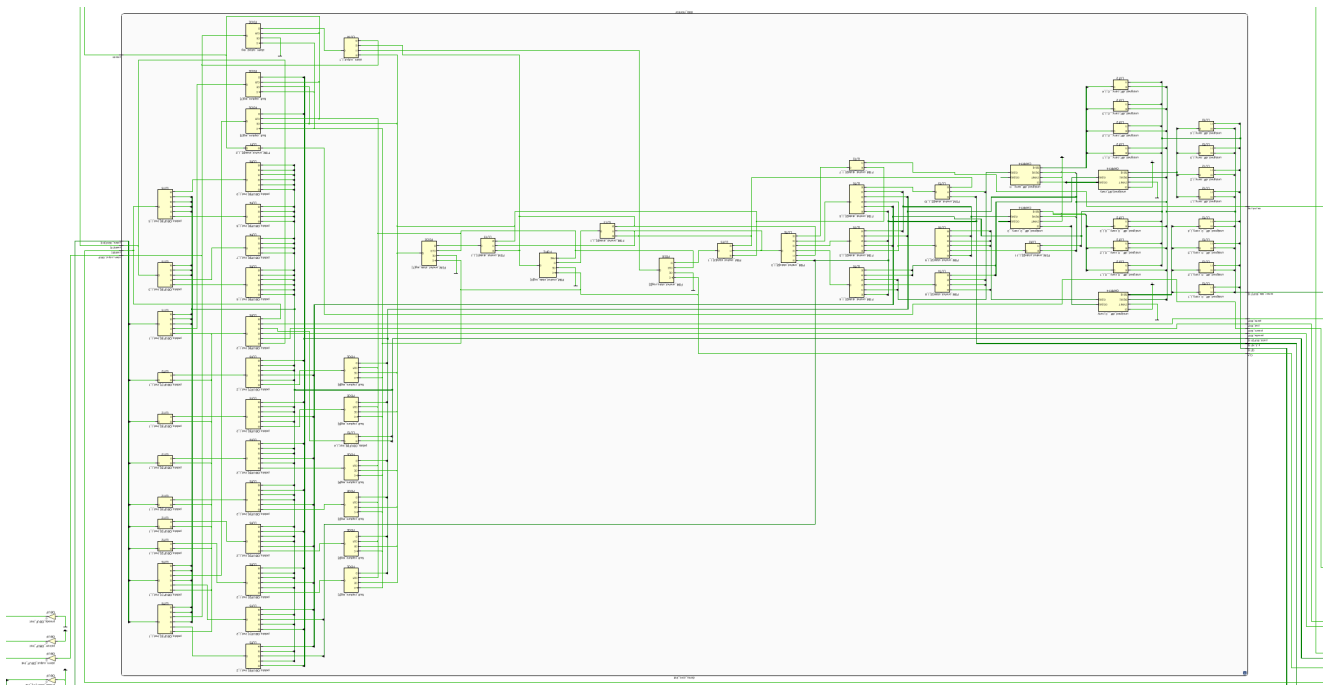


Fig. 3. Synthesized schematic – DEMU core and APB wrapper logic.

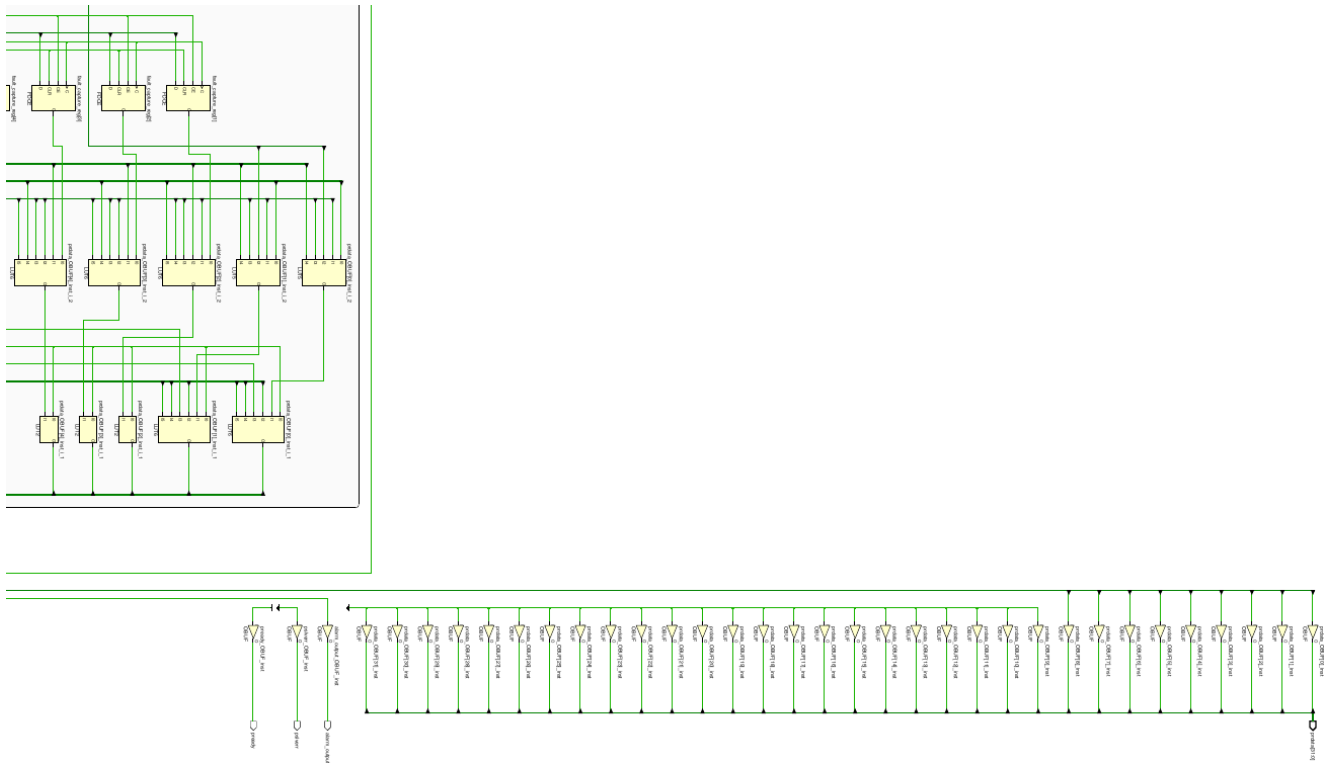


Fig. 4. Synthesized schematic – left-side I/O and mapped FPGA resources.

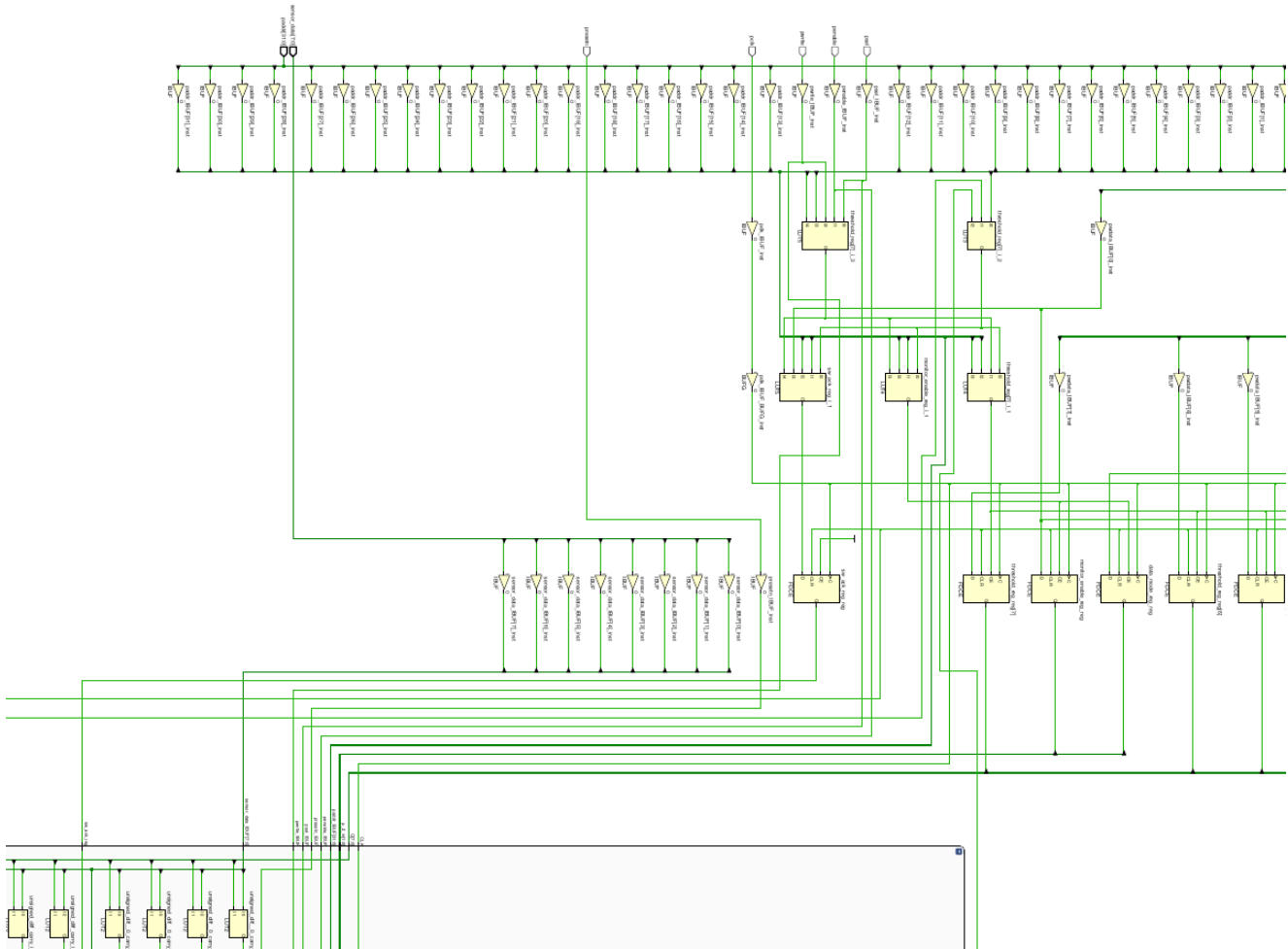


Fig. 5. Synthesized schematic – right-side I/O and mapped FPGA resources.

Name	Slice LUTs (14600)	Slice Registers (29200)	Bonded IOB (150)	BUFGCTRL (32)
apb_demu_wrapper	52	23	88	1
demu_core_inst(data_monitor)	47	12	0	0

Fig. 6. Post-synthesis utilization by hierarchy for apb_demu_wrapper.

Setup	Hold	Pulse Width
Worst Negative Slack (WNS): 5.022 ns	Worst Hold Slack (WHS): 0.143 ns	Worst Pulse Width Slack (WPWS): 4.500 ns
Total Negative Slack (TNS): 0.000 ns	Total Hold Slack (THS): 0.000 ns	Total Pulse Width Negative Slack (TPWS): 0.000 ns
Number of Failing Endpoints: 0	Number of Failing Endpoints: 0	Number of Failing Endpoints: 0
Total Number of Endpoints: 12	Total Number of Endpoints: 12	Total Number of Endpoints: 24
All user specified timing constraints are met.		

Fig. 7. Vivado timing summary after applying the 10 ns APB clock constraint.